

HARSHIT GANGWAR

B.Tech, Computer Science & Engineering | JSS Academy of Technical Education, Noida

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EDUCATION

B.Tech, Computer Science & Engineering — JSS Academy of Technical Education, Noida (JSSATE) 2024 - 2028
CPI: 8.42 / 10

Intermediate (+2), CBSE — KAMM Scholars School 2022 - 2024
Percentage: 76.8%

Matriculation, CBSE — KAMM Scholars School 2021 - 2022
Percentage: 98.00%

EXPERIENCE

Applied Data Science Lab / WorldQuant University Jan'25 - Dec'25

- Completed coursework in data ingestion, exploratory data analysis, regression, and logistic modeling as part of an applied data science program.
- Built end-to-end predictive models across real-world domains: housing price prediction (Argentina), air quality forecasting (Nairobi), bankruptcy prediction (Poland/Europe), earthquake damage prediction (Nepal), and market volatility forecasting on the BSE.

PROJECTS

AI Model for Bank Transaction Fraud Detection Jul'25 - Aug'25

- Trained a neural network from scratch on 9M+ bank transaction records to flag fraudulent transactions; achieved an 80% fraud detection (recall) rate, the highest reported on the Kaggle dataset leaderboard at the time. ([Kaggle Notebook](#))
- Performed in-depth exploratory analysis and feature engineering to handle severe class imbalance and improve model precision/recall trade-off.

Financial Market Analysis and Algorithmic Trading Apr'25 - May'25

- Built an end-to-end pipeline to analyze 10 years of Indian stock market data, engineer technical indicators (SMA, EMA, RSI, MACD), and design algorithmic trading strategies. ([GitHub Repo](#))
- Developed a custom backtesting engine to evaluate strategies against historical price data; SMA/EMA crossover strategy outperformed the buy-and-hold benchmark by 800x on Adani stock in simulation.

Vector Search Engine In Progress

- Implemented a brute-force k-nearest-neighbor search baseline, then an approximate nearest neighbor (ANN) algorithm achieving $O(n \log k)$ query complexity using a heap-based top-k selection.
- Currently implementing HNSW (Hierarchical Navigable Small World) graphs to achieve sub-linear approximate search over high-dimensional vector embeddings.

TECHNICAL SKILLS

Languages & Core: Java (Main), Python

Machine Learning / Data Science: Machine Learning, Neural Networks, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, seaborn, torchvision, torchaudio

Finance / Quant: yfinance, backtrader, portfolio optimization, algorithmic trading, backtesting

Tools: Git/GitHub, MS Office, Kaggle, Google Colab

ACHIEVEMENTS & EXTRACURRICULAR

- Solved 460+ problems on LeetCode across data structures and algorithms, with advanced proficiency in Dynamic Programming, Backtracking, and Union-Find.
- 2-Star rated on CodeChef.
- Core Team Member of the official Dance Society of the college.
- Reached the grand finale of the college-wide hackathon with the Food Connect project.